Sussex Centre for Consciousness Science

STROBOSCOPIC STIMULATION Standard Operating Procedure and Training Manual

GENERAL WORKING PROCEDURES
Chichester 2, Room 210

NEAREST FIRE ALARM: **Corridor**

NEAREST FIRE EXTINGUISHER: **By Lift**

NEAREST FIRST AID STATION: **By Lift**

WORKING WITH PARTICIPANTS

- *ILLNESS.* If either you or your participant is suspected of being ill, the research booking should be cancelled and, if possible, rescheduled. Please advise participants not to attend if they feel ill prior to the session.
- *FRAGRANCE-FREE.* You should ensure a fragrance-free environment for the comfort of participants since you will be in such close proximity. Avoid using strong scents on the day and of course be aware of personal hygiene!

WORKING ALONE

- *REDUCING RISK.* If you suspect you are working alone in the building, there are two things you can do to minimize risk:
    - 1) call security (01273 67 8234) to inform them of your location and how long you expect to be working
    - 2) call an independent party and inform them of your location and how long you expect to be working.
- Either party can then call to check on you during your time in the lab and you should remember to call relevant parties when you are leaving.

WEEKENDS. Give special consideration to these issues at the weekend, when it is more likely that you will be working alone

HANDLING MONEY

- *REDUCING RISK.* Make sure that the cash tin is always securely stored. Whenever possible, prepare payment without the participant present to maintain security, both of the cash tin and participants' paperwork.

EMERGENCY PROCEDURES

Make sure you are aware of the nearest landline to use in case of an emergency. In the EEGlab (room 210) there is a phone located under the TMS within the testing room

IN ALL CASES OF DOUBT OR WHEN SYMPTOMS PERSIST, ALWAYS SEEK MEDICAL ATTENTION

What to do in the Case of an Adverse Reaction

In the case of an adverse reaction follow these steps:

1. Assess the participant immediately based on your *Emergency first aid at work* training. This will include responsiveness, awareness and brief questioning about their situation.
2. If anything is highlighted by this questioning, call the Sussex Centre for Consciousness Science lab manager David Schwartzman on Tel: 07709918923.
3. If necessary, the lab manager will call security 01273 (87) 3333.
4. If you are unable to reach the lab manager, if you deem it necessary you will need to **call security 01273 (87) 3333**, who will guide the ambulance to your location.

To minimise the risk of adverse reaction, during the experiment keep constant communication between you and the participant. This allows you to monitor for any adverse side effects and for the participant to indicate their level of comfort or ask questions at any point during the testing.

After each experimental condition in which any stroboscopic light is administered and throughout the experiment the researcher should constantly assess the participant to make sure they are OK.

Finally, if you are also recording the participant's EEG, you have a direct window into their brain activity, which can be used to detect the onset of any potential seizure activity, under which circumstances **stop the experiment immediately**.

Following completion of the experimental sessions, while the participant washes their hair and is debriefed about the experiment, you can monitor for any adverse side effects.

If any adverse reaction occurs, following assessment by the lab manager, if they deem the participant to be fit and well, you should encourage the participant to stay with you for an hour to make sure that they do not display any further reactions. You should assess the participant throughout this time and if anything is highlighted, follow the procedure above. If after 1 hour has passed and the participant shows no additional symptoms, they will be free to leave. You should make the participant aware of the reasons behind their reaction (seizure) and advise them to seek a medical consultation to investigate their condition further.

If you have any questions about best conduct or procedures, please contact the lab manager: D.schwartzman@sussex.ac.uk Tel: +44-1273-67-8213

EXCLUSION CRITERIA

Before participants are allowed to take part in any study involving stroboscopic stimulation they must complete:

1. Safety Screening Questionnaire for Stroboscopic Studies
2. Anxiety Screening Questionnaire

Safety screening: Individuals with a history of epilepsy highlighted by the screening questionnaire (see below) must be excluded from participating in the study. If participants answer **YES** to **ANY** of the questions in the safety screening they must be excluded from participation.

Migraine question: Flickering lights may trigger the onset of Migraines in some people. Therefore, anyone with a history of Migraines must also be excluded from participation. In the introductory email the researcher can ask the following question: "Do you frequently suffer from migraines or regular headaches", if the participant indicates that they do, they must be excluded from participation.

Anxiety questionnaire: Flickering lights have been shown to cause mild anxiety in a small percentage of the general population. To guard against this concern participants must also complete the State-Trait Anxiety Index (STAI, see attached), Trait Version before taking part. All items are rated on a 4-point scale (e.g., from "Almost Never" to "Almost Always"). The STAI yields a total score between 20 and 80, with values above 50 indicating moderate anxiety. Participants reporting scores above 50 must be excluded from participation.

For more information see Spielberger, C. D., Gorsuch, R. L., Lushene, R., Vagg, P. R., & Jacobs, G. A. (1983). *Manual for the State-Trait Anxiety Inventory*. Palo Alto, CA: Consulting Psychologists Press.

STANDARD OPERATING PROCEDURES

In order to run an experiment with Lucia or any stroboscopic light in the EEG lab, proof of current first aid responder training is required.

BEFORE SESSION

- Participant should have been sent exclusion criteria and questionnaires via email before attending
- Participant should wear layered clothing.
- Has participant been to bathroom?
- Turn on the infrared camera; this must be on during the experiment to allow you to closely monitor the participant throughout the experiment.
- Photobiological safety – it is up to you to make sure that any experiment you are running using stroboscopic stimulation does not exceed a maximum luminance of 7000 lux, adhering to European directive guidelines (Directive-2006/25/EC).

ONCE PARTICIPANT ARRIVES

1. Read out exclusion criteria questionnaire again and get them to sign it.

2. Always run a brief stroboscopic taster session with the participant before beginning the experiment to identify any potential adverse reactions or aversion to stroboscopic lighting.

<u>If they answer yes to any of the questions, they may NOT take part in the study</u>

University of Sussex

Safety Screening Questionnaire for Stroboscopic Studies

Do any of the following apply to you? **Please enter a 'X' to signify Yes or No for each question.** Please let the researcher know if you are unclear about any of these statements.

	Yes	No
1. Are you 18 years of age or older?	☐	☐
2. Do you find bright or flashing lights unpleasant?	☐	☐
3. Do you have epilepsy, or have you ever been told that you do?	☐	☐
4. Has anyone in your immediate biological family been told they have photosensitive epilepsy?	☐	☐
5. In the past three years, have you had an episode of psychosis or a severe anxiety disorder?	☐	☐
6. Do you frequently have headaches or migraines which may be triggered by bright lights?	☐	☐
7. Have you ever had repeated episodes in which you suddenly lost consciousness, which were unexplained?	☐	☐
8. Are you currently pregnant?	☐	☐

Name: _____

Signature: _____

Date: .....

Non-Safety Related Questions:

What is your level of proficiency with the English language?

Native speaker ☐ Fluent ☐ Advanced ☐ Intermediate ☐ Beginner ☐

Do you have any untreated or untreatable problems with your vision? This could mean partial blindness, colour blindness, or blurry vision that has not been treated with glasses.

☐ Yes, I have problems with my vision.

☐ No, I have no problems with my vision.

Anxiety Screening Questionnaire (STAI)

Anxiety Screening Questionnaire

	Almost never	Sometimes	Often	Almost Always
I feel pleasant.				
I feel nervous and restless.				
I feel satisfied with myself.				
I wish I could be as happy others seem to be.				
I feel like a failure.				
I feel rested.				
I am 'calm, cool and collected'.				
	Almost never	**Sometimes**	**Often**	**Almost Always**
I feel that difficulties are piling up so I cannot overcome them.				
I worry too much over something that doesn't really matter.				
I am happy.				
I have disturbing thoughts.				
I lack self-confidence.				
I feel secure.				
I make decisions easily.				
	Almost never	**Sometimes**	**Often**	**Almost Always**
I feel inadequate.				
I am content.				
Some unimportant thought runs through my mind and bothers me.				
I take disappointments so keenly that I can't put them out of my mind.				
I am a steady person.				
I get in a state of tension or turmoil as I think over my recent concerns and interests.				

CONFIRMATION

STROBOSCOPIC STIMULATION TRAINING MANUAL

NAME: _____

I CONFIRM THAT I HAVE READ, UNDERSTOOD AND, WHERE APPROPRIATE, SIGNED THE FOLLOWING DOCUMENTS:

STROBOSCOPIC TRAINING MANUAL _____

SIGNATURE: _____ DATE: _____

Please email a signed copy of this page to the lab manager: d.schwartzman@sussex.ac.uk